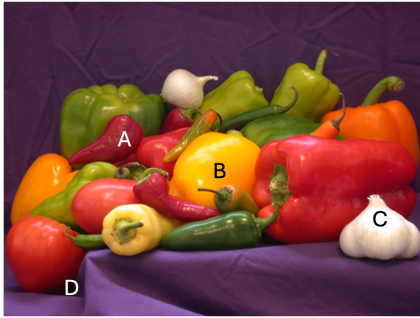


Question 1: Colour Representations & Greyscale Conversions (4 pts: $0.5 \times 4 + 1 + 1$)



Colour Image



Greyscale Version 1



Greyscale Version 2

Consider the locations at the center of the letters in the colour image. For Location A, what is the RGB value?

- A. [147, 34, 52]
- B. [254, 254, 254]
- C. [16, 8, 12]
- D. [255, 201, 18]
- E. [112, 111, 58]

Consider the locations at the center of the letters in the colour image. For location B, what is the normalized RGB value?

- A. [0.63, 0.15, 78]
- B. [0.54, 0.42, 158]
- C. [0.33, 0.33, 254]
- D. [0.44, 0.22, 12]
- E. [0.40, 0.40, 40]

Consider the locations at the center of the letters in the colour image. For location C, what is the V of the HSV value?

- A. 147; B. 254; C. 16; D. 255; E. 112

Consider the locations at the center of the letters in the colour image. For location D, what is the intensity?

- A. 78, B. 58, C. 254, D. 12, E. 40

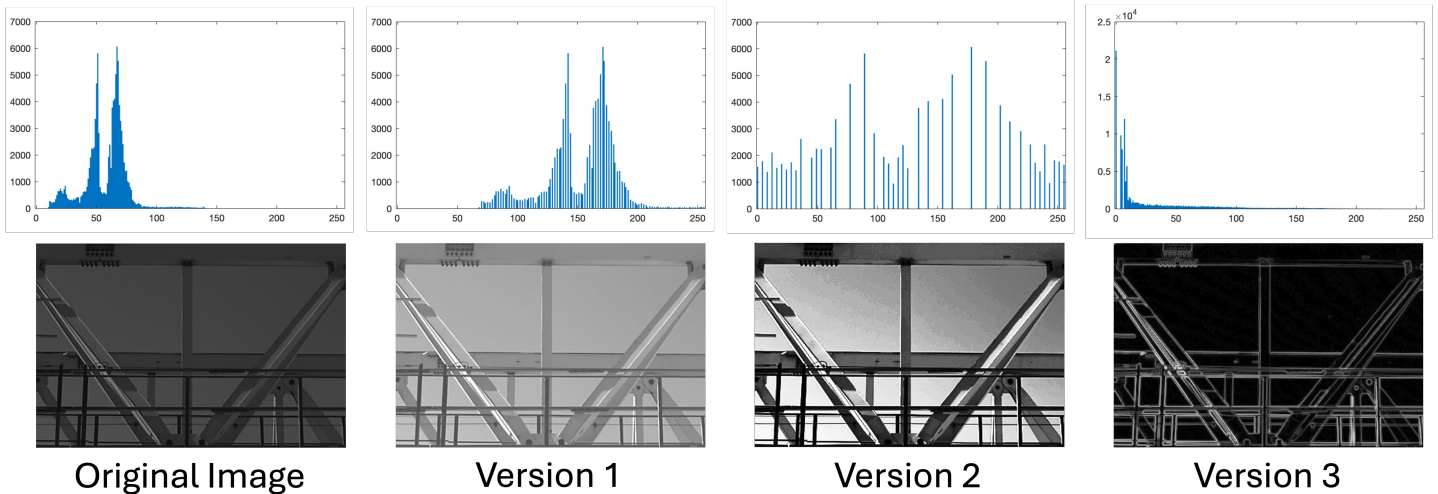
Consider the locations at the center of the letters in the colour image. Which weighting $[W_R, W_G, W_B]$ was used to convert the colour image to greyscale image version 1?

- A. [0.950, 0.025, 0.025]; B. [0.025, 0.950, 0.025]
- C. [0.025, 0.025, 0.950]; D. [0.299, 0.587, 0.114]
- E. [0.33, 0.33, 0.33]

Which weighting $[W_R, W_G, W_B]$ was used to convert the colour image to greyscale image version 2?

- A. [0.950, 0.025, 0.025]; B. [0.025, 0.950, 0.025]
- C. [0.025, 0.025, 0.950]; D. [0.299, 0.587, 0.114]
- E. [0.33, 0.33, 0.33]

Question 2: Image Histograms (5 pts; 3 + 1 + 1)



Consider the original image on the left and three transformed versions of it. Each image's associated intensity histograms is plotted above. Select the appropriate transformation that leads to each of the three versions.

Version 1

- A. Brightness Adjustment + Intensity Scaling
- B. Gamma Mapping
- C. Histogram Stretching
- D. Histogram Equalization
- E. Image Gradient Magnitude
- F. Canny Edge Detection
- G. Median Filtering
- H. None of the other options

Version 2

- A. Brightness Adjustment + Intensity Scaling
- B. Gamma Mapping
- C. Histogram Stretching
- D. Histogram Equalization
- E. Image Gradient Magnitude
- F. Canny Edge Detection
- G. Median Filtering
- H. None of the other options

Version 3

- A. Brightness Adjustment + Intensity Scaling
- B. Gamma Mapping
- C. Histogram Stretching
- D. Histogram Equalization
- E. Image Gradient Magnitude
- F. Canny Edge Detection
- G. Median Filtering
- H. None of the other options

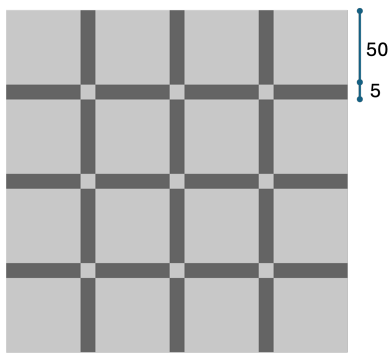
You apply Otsu's method to the images to try and separate the foreground (metal structures) from the background (sky). Will applying Otsu's method to transformed Version 1 lead to the same foreground-background segmentation as the original?

Yes / No / Can't tell

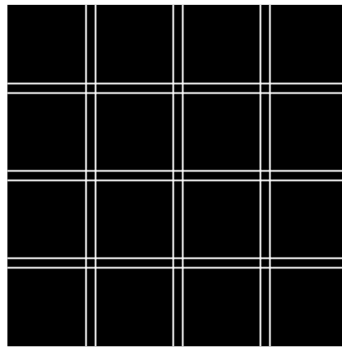
Will apply Otsu's method to transformed Version 2 lead to the same foreground-background segmentation as the original? Why or why not?

No / No / Can't tell

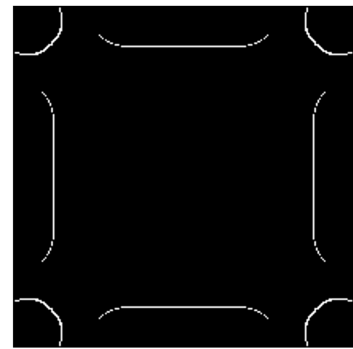
Question 3: Filtering + Canny (6 pts + 3pts)



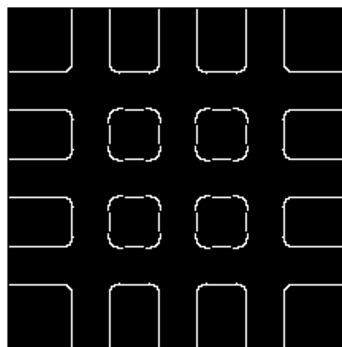
Original Image



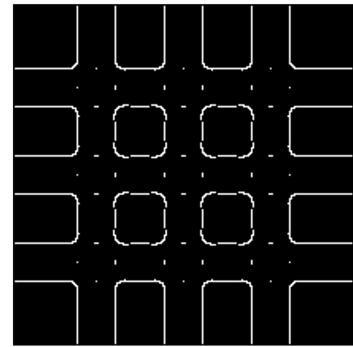
Canny Result 1



Canny Result 2



Canny Result 3



Canny Result 4

In the original image above, the large squares are 50x50 of intensity 200, while the bars are of height 5, with intensity 100.

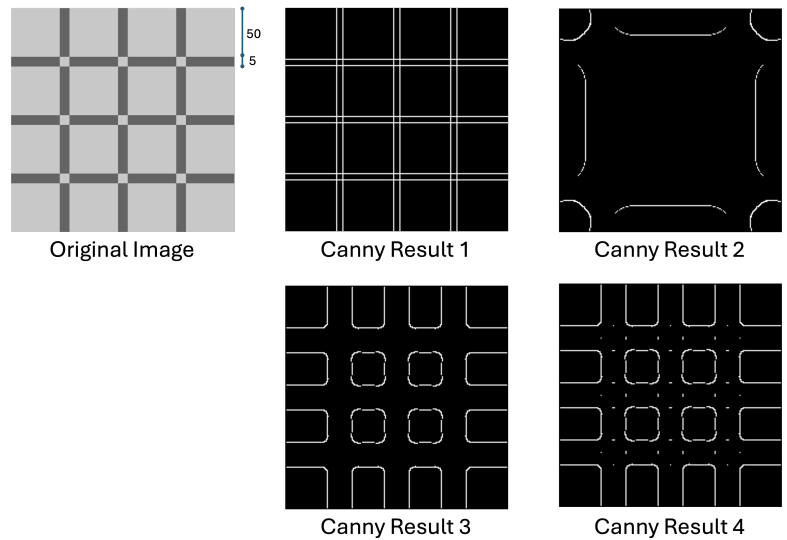
Consider convolving the original image with the following filters. Do not pad borders. When applying normalized filters, convert outputs back to integers between 0-255. How many unique intensity values will there be when convolving with a

- (a) a horizontal Sobel filter: ___ unique value(s)
- (b) a 2D Laplace filter $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$: ___ unique value(s)
- (c) a normalized Gaussian filter of 3x3: ___ unique value(s)
- (d) a normalized box filter of 55x55: ___ unique value(s)

Consider that the original image is corrupted with noise. You opt to apply a 3x3 median filter. Assume that the noise is spatially distributed in a uniformly random manner, but each noise pixel does not have any other noise pixels in its 8-neighbour vicinity. How many unique intensity values will there be if

- (e) the noise is salt-and-pepper noise: ___ unique values
- (f) the noise is additive Gaussian noise, drawn from a normal distribution $\mathcal{N}(0, 20^2)$:
___ unique values

Question 3 continued



You run Canny edge detection on the original image. Depending on the parameter settings, you get 4 different results shown above. Consider the following pairs of Canny results and specify the parameter settings that lead to the different results.

Canny results 1 and 3:

- A. The sigma of the Gaussian filter for Result 1 is smaller than Result 3
- B. The sigma of the Gaussian filter for Result 1 is larger than Result 3
- C. The gradient magnitude threshold for Result 1 is set lower than Result 3
- D. The gradient magnitude threshold for Result 1 is set higher than Result 3
- E. The high threshold of hysteresis thresholding of Result 1 is set lower than Result 3
- F. The high threshold of hysteresis thresholding of Result 1 is set higher than Result 3
- G. Result 1 uses gradient interpolation during non-maximum suppression whereas Result 3 uses the nearest pixel approximation

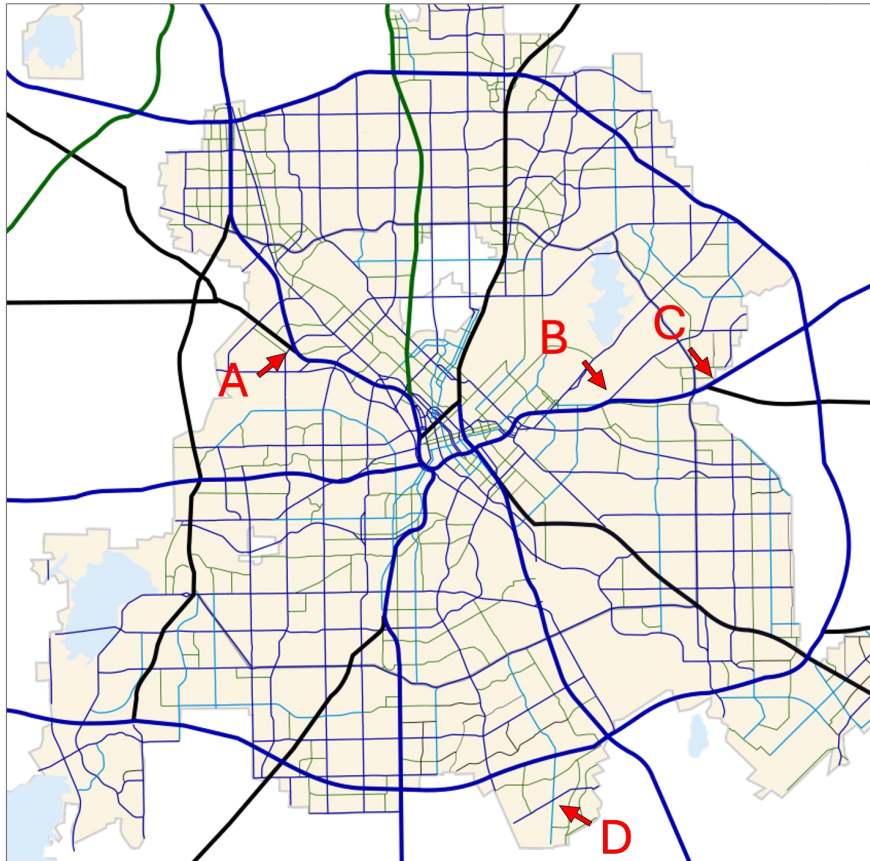
Canny results 2 and 3:

- A. The sigma of the Gaussian filter for Result 2 is smaller than Result 3
- B. The sigma of the Gaussian filter for Result 2 is larger than Result 3
- C. The gradient magnitude threshold for Result 2 is set lower than Result 3
- D. The gradient magnitude threshold for Result 2 is set higher than Result 3
- E. The high threshold of hysteresis thresholding of Result 2 is set lower than Result 3
- F. The high threshold of hysteresis thresholding of Result 2 is set higher than Result 3
- G. Result 2 uses gradient interpolation during non-maximum suppression whereas Result 3 uses the nearest pixel approximation

Canny results 3 and 4:

- A. The sigma of the Gaussian filter for Result 3 is smaller than Result 4
- B. The sigma of the Gaussian filter for Result 3 is larger than Result 4
- C. The gradient magnitude threshold for Result 3 is set lower than Result 4
- D. The gradient magnitude threshold for Result 3 is set higher than Result 4
- E. The high threshold of hysteresis thresholding of Result 3 is set lower than Result 4
- F. The high threshold of hysteresis thresholding of Result 3 is set higher than Result 4
- G. Result 3 uses gradient interpolation during non-maximum suppression whereas Result r uses the nearest pixel approximation

Question 4: System Design (6 points)



Dallas Road Map

Consider the above road map of the city of Texas. There are three types of roads: highways (thick blue, thick black lines), regular (dark blue thin lines) and minor (light blue, green thin lines). You are tasked with finding all the intersections which range between 30 and 60 degrees. Some examples are marked by red arrows in the above map.

- (a) Outline key steps of a method to find the intersections using the Hough transform. State any assumptions made for the solution.
- (b) For the Hough voting space, consider the coarsest angular quantization. How many bins are there?
- (c) Assume that you wish to find only the 30-50 degree intersections with highways (marked examples A and C) and exclude those in which one (B) or both roads (D) are regular or minor. What changes would you make to your method given in (a)?
- (d) Assume that you wish to find the 30-50 degree intersections as long as one of the roads are not light blue minor roads (marked example D). What changes would you make to your method given in (a).